AQA 3.1.9 Rate equations

AQA A-level Chemistry · Revision questions · Sheet 2 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 2 of 6

20 marks · about 25 minutes

Key information

- rate = $k[A]^m[B]^n$ — orders come from experimental data, not from the balanced equation
- Doubling a concentration: rate × 1 (zero order), × 2 (first order), × 4 (second order)
- Units of k: rearrange k = rate ÷ (concentration terms), with rate in mol dm^{-3} s^{-1}, and cancel
- Rate-determining step = slowest step; species in the rate equation appear in (or before) it
- $\ln k = -E_a/RT + \ln A$, so the gradient of ln k against 1/T = $-E_a/R$

Section A: Short-answer questions

01 The rate equation for a reaction is rate = k[X][Z]. State what this rate equation shows about the effect of the concentration of a third reactant, Y, on the rate of reaction.

[1 mark]

02 The rate equation for a reaction is rate = $k[C]^2[D]$. When [C] = 0.020 mol dm^{-3} and [D] = 0.15 mol dm^{-3} the initial rate is 4.8×10^{-4} mol dm^{-3} s^{-1}. Calculate the value of k and give its units.

[2 marks]

03 The rate constant k for the reaction of a bromoalkane with KOH is determined at a series of temperatures T. One result is shown: T = 52.2 °C, k = 1.65 mol^{-1} dm^3 s^{-1}, ln k = 0.50

Calculate the value of 1/T, in K^{-1}, for this result.

[1 mark]

04 State and explain the effect of adding a catalyst on the value of the rate constant, k, for a reaction.

[2 marks]

05 The rate equation for a reaction is rate = $k[H_2][NO]^2$. Give the overall order of the reaction and the units of k.

[2 marks]

Section B: Longer questions

06 The table shows the results of experiments on a reaction between A, B and C at a constant temperature.

Experiment	[A] / mol dm^{-3}	[B] / mol dm^{-3}	[C] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.10	0.10	0.20	2.0×10^{-4}
2	0.20	0.10	0.20	8.0×10^{-4}
3	0.10	0.30	0.20	6.0×10^{-4}
4	0.10	0.10	0.40	2.0×10^{-4}
5	0.30	0.20	0.10	

06.1 Deduce the order of reaction with respect to each of A, B and C.

[3 marks]

06.2 Give the rate equation for the reaction.

[1 mark]

06.3 Use the data from Experiment 1 to calculate the value of the rate constant, k, and give its units.

[2 marks]

06.4 Calculate the initial rate in Experiment 5.

[1 mark]

07 A and B react together in the presence of an acid catalyst.

$A(aq) + 2B(aq) \rightarrow C(aq) + D(aq)$

The rate equation for this reaction is rate = $k[B]^2[H^+]$

Table 1 shows how the values of the relative initial rate vary with different concentrations of each reagent at the same temperature.

Experiment	[A] / mol dm^{-3}	[B] / mol dm^{-3}	$[H^+]$ / mol dm^{-3}	Relative initial rate
1	0.40	0.20	0.10	1.00
2	0.50	0.20	0.10	
3	0.40		0.10	0.64
4	0.50	0.30	0.06	

07.1 Complete Table 1 by calculating the missing values.

[3 marks]

07.2 A suggested mechanism for the reaction is shown.

Step 1 $B + H^+ \rightarrow BH^+$

Step 2 $BH^+ + B \rightarrow B_2H^+$

Step 3 $B_2H^+ + A \rightarrow C + D$

Deduce the rate-determining step for this reaction. Give a reason for your answer.

[2 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	Y is zero order / changing [Y] has no effect on the rate	*Source: Practice question (written for this set)*	1
02	M1 $k = 4.8 \times 10^{-4} \div (0.020^2 \times 0.15) = 8.0$ M2 $mol^{-2}\ dm^6\ s^{-1}$	*Source: Practice question (written for this set)*	2
03	3.08×10^{-3}	*Source: 7405/2 Jun 2025 Q09.3* Allow 0.003075 or 3.075×10^{-3} or 0.00308 Original asks students to complete the missing entry in a results table	1
04	M1 k increases M2 catalyst provides an alternative route with a lower activation energy	*Source: Practice question (written for this set)*	2
05	third order / 3 $mol^{-2}\ dm^6\ s^{-1}$	*Source: Practice question (written for this set)*	2
06.1	A second order B first order C zero order	*Source: Practice question (written for this set)*	3
06.2	rate = $k[A]^2[B]$		1
06.3	$k = 2.0 \times 10^{-4} \div (0.10^2 \times 0.10) = 0.20$ $mol^{-2}\ dm^6\ s^{-1}$	Allow ECF from their rate equation	2
06.4	3.6×10^{-3} mol $dm^{-3}\ s^{-1}$	ECF from their k and rate equation	1
07.1	M1 Experiment 2 relative rate = 1.00 M2 Experiment 3 [B] = 0.16 M3 Experiment 4 relative rate = 1.35	*Source: 7405/2 Jun 2023 Q02*	3
07.2	M1 Step 2 M2 (by the end of step 2) 1 × H^+ and 2 × B have been used		2